

Ethan Huber

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Technical Skills

Programming Languages: C, VHDL, Go, Java, C#, TypeScript, JavaScript, Python, HTML/CSS
Systems & Protocols: STM32, ESP32, FPGA, Raspberry Pi, UART, I2C, CAN Bus
Tools & Infrastructure: Git, Linux, Docker, Kafka, PostgreSQL, MariaDB, SQLite, REST API, Angular, .NET
Design & Manufacturing: OnShape, KiCad, Altium, Solidworks, Vivado, Xilinx ISE, 3D Printing, SMD Soldering

Work Experience

Systems Engineering Co-op Analyst

May 2026 – Aug 2026

Toyota Motor Manufacturing Canada

Cambridge, Ontario

- Co-led the design and deployment of a **Python** ML anomaly detection system, containerized with **Docker**, ingesting live torque controller data through **Kafka** into **PostgreSQL** for real-time manufacturing defect detection
- Reverse engineered a proprietary torque controller protocol and built a custom software adapter for live data capture, partnering with assembly technicians and engineers to gather requirements
- Developed a full-stack test automation framework (**TypeScript**, **Playwright**) to streamline regression testing across 254 features enabling product owners to manage testing scope prioritization efficiently
- Contributed to an AGV traffic control system with **Angular** and **.NET**, following **Git** and **Agile** workflows

Process Control Team Member

May 2025 – May 2026

Toyota Motor Manufacturing Canada

Cambridge, Ontario

- Supported final assembly operations using Toyota Production System principles to maintain workflow efficiency
- Identified 19 defects during vehicle shipping preparation, preventing quality issues before customer delivery

Education

Bachelor of Engineering, Computer Engineering (Co-op)

Sept 2024 – May 2029 (Expected)

University of Guelph; 87% average (GPA: 3.7)

Guelph, Ontario

- Relevant courses: Data Structures and Algorithms, Engineering Systems Analysis, Object Oriented Programming for Engineers, Digital System Design, Software Systems Development and Integration, Computer Organization

Extracurricular Experience

Electrical Team Lead

Sep 2025 – Present

University of Guelph Robotics Team

Promoted to Lead in May 2026

- Helped lead the electrical sub-team, coordinating with mechanical/software teams to develop, test, and integrate rovers electrical systems from power distribution to the CAN bus for the CIRC
- Designed and routed multiple PCBs in KiCad, optimizing trace sizing, connector placement, and component layout
- Validated system performance using oscilloscopes and multimeters (debugging CAN, verifying power regulation across 24V/12V/5V rails) and contributed to safety documentation for competition compliance

Projects

Modular Agricultural Robot Platform // April 2026 – Present

github.com/zushiEdu/GRP2 🔗

- Engineered a repairable agricultural robot platform with a sheet metal aluminum chassis, custom 3D printed in-hub gearboxes, modular sensor bumpers, and a swappable payload mount for easy repair and upgrade
- Designed and built a custom **STM32** control board with 4 MOSFET motor drivers for real time motor control, communicating with a Raspberry Pi over **UART** and supporting **I2C** for modular tool and sensor expansion
- Built an **Angular** web based mission manager on a base station for remote monitoring and control of robots
- Currently developing GPS + IMU guided autonomous navigation with computer vision on the Raspberry Pi, aiming to enable the robot to scout corn fields and return data autonomously

Filedex - Semantic file search engine // Jul 2025 – Sep 2025

github.com/zushiEdu/filedex 🔗

- Developed a semantic file search engine in **Go** for file indexing and retrieval, using **SQLite** for metadata storage
- Implemented graph traversal algorithms to identify relationships between tagged files
- Deployed local AI/ML recognition using Ollama to automate metadata tagging generation

Basic Messaging App // May 2025 – Jul 2025

github.com/zushiEdu/basicmessenger 🔗

- Built a messaging web app with a **Go REST API** backed by **MariaDB**, serving a HTML/CSS/JavaScript frontend
- Containerized the backend in **Docker** and deployed to a self built Linux **Unraid** server with **GitHub Actions** CI/CD auto building images on every commit so I never have to manually update the container